

GANESHAN GOWTHAN

MECHANICAL / MECHATRONICS ENGINEER

 ganeshangowthan1998@gmail.com

 [Ganeshan Gowthan](#)

 +94 76 220 6091

DESIGN PORTFOLIO

CREATED: 03/10/2024

LAST MODIFIED: 05/06/2026

DESIGN AND DEVELOPMENT OF THE KASPER FLYING OBJECT (KFO)

KASPER GLOBAL (PVT) LTD.

[CLICK HERE TO VISIT WEBSITE.](#)

Overview:

The Kasper Flying Object (KFO) is a flagship product of Kasper Global (Pvt) Ltd., which is a presence detector integrated with IR, Temperature Control, Lighting Control and Bluetooth. The previous version was bulky and intrusive. This project was focused on the re-design and development of a sleek, less intrusive version of the KFO.

Objectives and Aims:

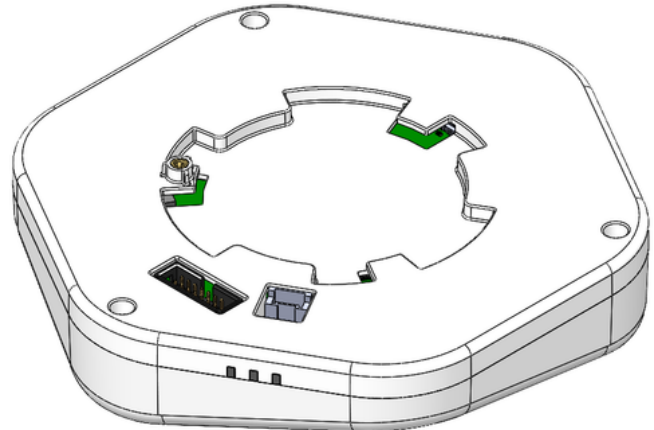
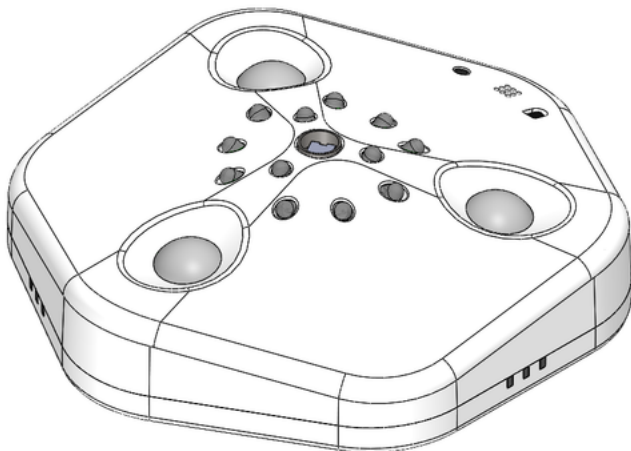
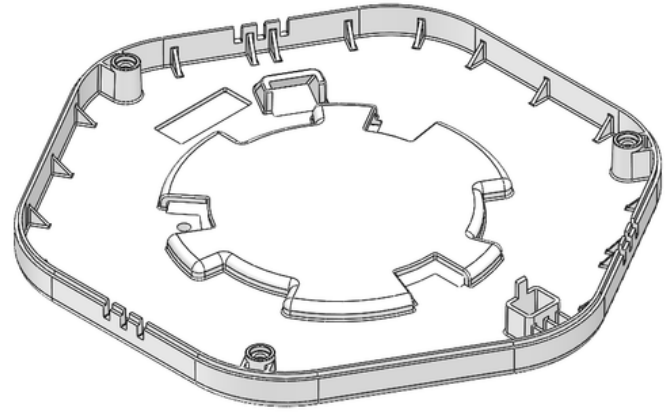
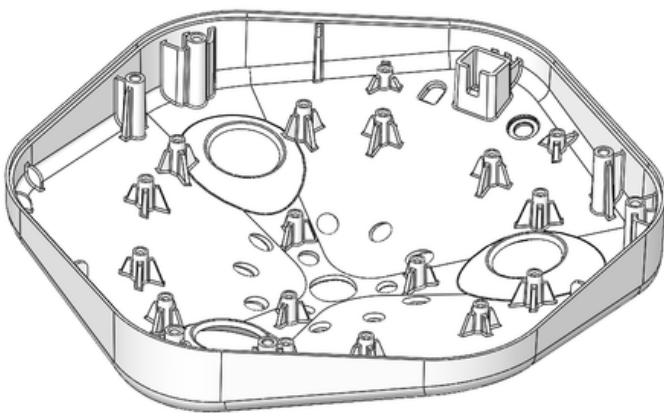
- Enhance aesthetics and ergonomics
- Perform DFMA with mass manufacturing taken into account
- Ensure easy assembly and disassembly

Material Selection:

- ABS + PC for strength and durability

Prototyping Methods

- FDM 3D Printing using PLA and ABS



DESIGN AND DEVELOPMENT OF AN ELECTRIC MOTORCYCLE FRAME

THERMALR INDUSTRIES (PVT) LTD.



[CLICK HERE TO VISIT WEBSITE.](#)

Overview:

This project was focused on the re-design and development of a lightweight, durable frame for an electric motorcycle, optimizing performance, aesthetics, and structural integrity. The initial motorcycle frame had issues such as larger wheelbase (1400mm), higher seat height, asymmetry, etc.

Objectives and Aims:

- Maximize rigidity while minimizing weight
- Enhance aesthetics and ergonomics
- Ensure easy assembly and maintenance
- Ensure a safe design during worst case loading conditions i.e. maintaining an acceptable Safety Factor value during accelerating, braking and cornering

Material Selection:

- Q195 Mild Steel was used to achieve a balance of strength and weight

Outcomes:

- Successfully developed a trellis frame and achieved significant weight reduction while enhancing stiffness.
- Received positive feedback for the frame's performance, aesthetics and ergonomics

Design Process:

- Designed using SolidWorks Weldments Tool
- Conducted FEA using ANSYS for static and dynamic loading to optimize geometry.

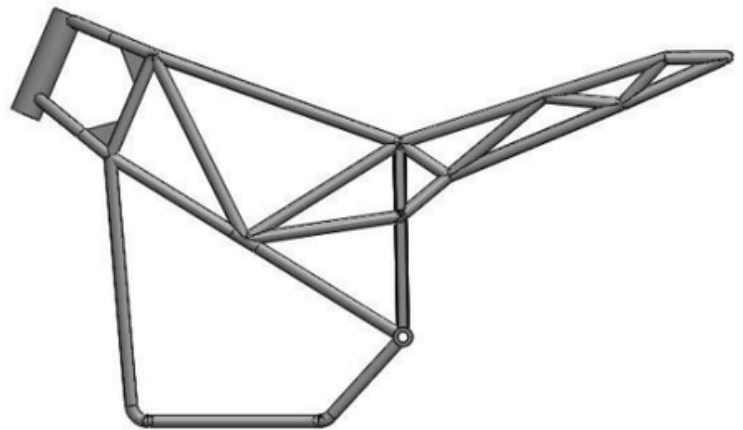


Figure 1: Initial Frame Design - 18.5kg



Figure 2: Final Frame Design - 12.6kg

DESIGN AND MANUFACTURING OF A CUSTOM WELDING JIG FOR MOTORCYCLE FRAME MANUFACTURING

THERMALR INDUSTRIES (PVT) LTD.



[CLICK HERE TO VISIT WEBSITE.](#)

Overview:

This project focused on the design and manufacturing of a custom welding jig specifically for assembling a trellis frame for an electric motorcycle. The goal was to enhance precision, efficiency, and repeatability during the welding process.

Objectives and Aims:

- Ensure accurate alignment of frame components.
- Facilitate easy access for welding without compromising stability.

Material Selection:

- Q195 Mild Steel Box Bars and 1mm thick Mild Steel Sheets

Design Process:

- Designed using SolidWorks Weldments and Sheet Metal Tools
- Conducted FEA using ANSYS to check for structural integrity.

Outcomes:

- Enhanced the overall quality of the motorcycle frame with a few defects and rework

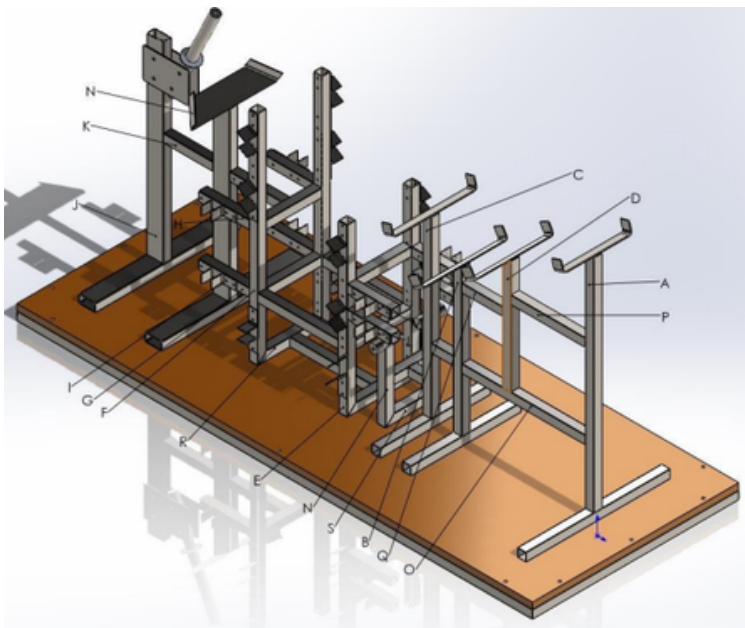


Figure 3: Final Design of the Custom Welding Jig

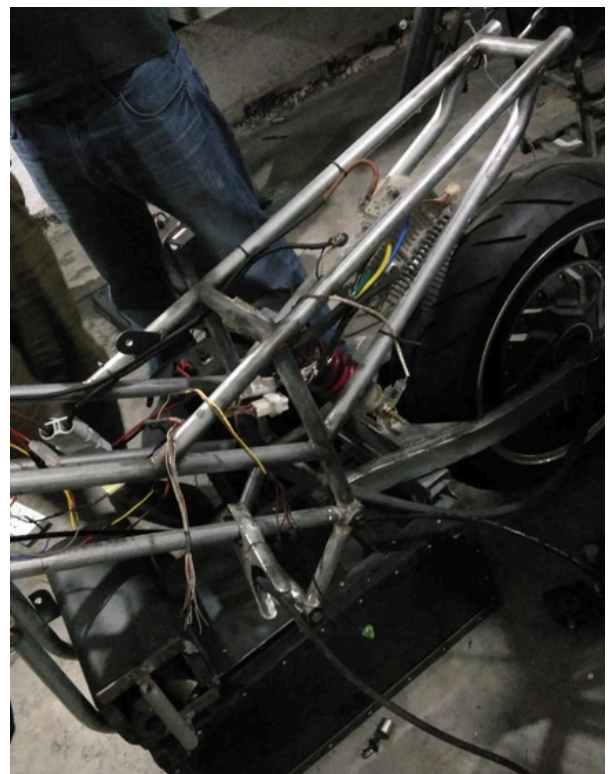


Figure 4: Manufactured Motorcycle Frame

DESIGN OF BODY PANELS FOR THE ELECTRIC MOTORCYCLE PROJECT

THERMALR INDUSTRIES (PVT) LTD.



[CLICK HERE TO VISIT WEBSITE.](#)

Overview:

This project focused on the design and prototyping of motorcycle body panels using surface modelling techniques in SolidWorks, aiming to enhance aesthetics and manufacturability.

Objectives and Aims:

- Create visually appealing and aerodynamic body panels.
- Facilitate easy assembly and maintenance.

Design Process:

- Utilized 2D and 3D Sketching, Projected Curves, Boundary Surfacing, Surface Lofts, etc.
- Rendered using SolidWorks Visualize
- Collaborated with my team mates to integrate stylistic elements.

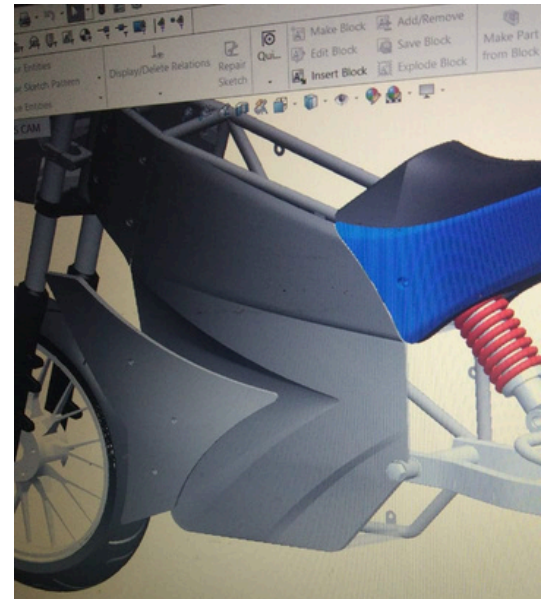


Figure 5: Design and Assembly

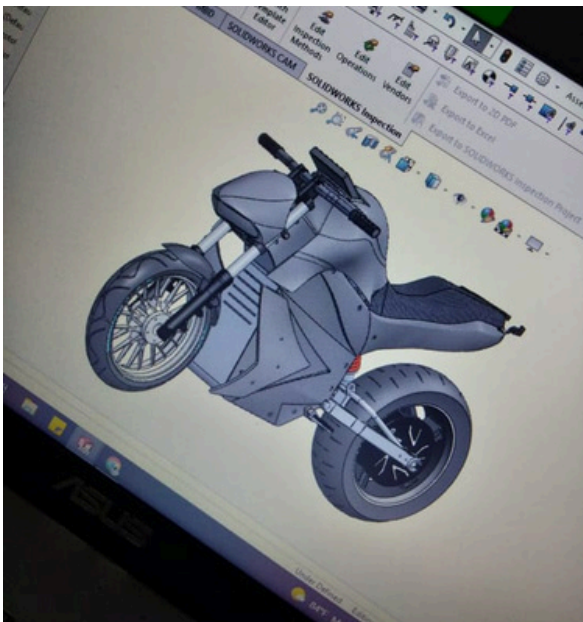


Figure 6: Finalized Model



Figure 7: Rendered Model

By SHANUKA KADUPITTIYAGE

The falcon has never failed to capture the imagination of man. It is a bird that is fast, precise and lethal. It is a species of bird that has captivated man, symbolising freedom, power, victory and ambition. It is also the source of inspiration behind Sri Lanka's very own high-performance electric motorcycle, The Peregrine by Thermalit.

Recently unveiled at the national industrial exhibition, Industry 2022, Thermalit's all-new, all-electric motorcycle became a viral sensation in the country overnight, even making headlines. Ceylon Today reached out to Founder and CEO of Thermalit Industries, Sasiranga De Silva, curious to learn more.

Meet Sasiranga De Silva

You may have heard the name Sasiranga De Silva before. He is the same person that made headlines a couple of years ago for creating an electric conversion kit for tuktuks, an achievement that earned him multiple awards and accolades internationally.

"I've always been a massive fan of automobiles and vehicles growing up," Sasiranga shared while speaking with Ceylon Today. "But I was also conscious of the environmental impact that it caused as a result of burning fossil fuels."

As Sasiranga studied more about environmentally friendly products and automobiles, his interest directed towards electric vehicles, which had no emissions and could run on clean energy.

"I felt electric vehicles was where I should be doing research and development, which I did after becoming a lecturer at the University of Moratuwa around April, 2012."

In between his lectures, Sasiranga was involved in research on electric vehicles, which led to the creation of the electric tuk conversion kit many years later.

Schubbing his innovation to the UN Environment's Asia-Pacific Low Carbon Lifestyles Challenge, Sasiranga ended up winning the award, which also earned him a US\$20,000 grant as well as international recognition.

Explaining the inspiration behind his creations, Sasiranga explained that "one of the main sources of pollution in Sri Lanka are the vehicles on its roads, and the tuktuk is one of the most common vehicles found on Sri Lankan roads. Just imagine how much pollution could be avoided if we were able to convert these vehicles into electric power," he explained enthusiastically.

From electric tuk to motorcycles

Sadly, Sasiranga's view wasn't mirrored within Sri Lanka's borders. The initial investment needed for the conversion turned out to be too steep of an asking price thanks to the high quality lithium ion batteries Sasiranga uses for his conversions. That, along with the lack of awareness on EVs and their benefits among the tuk community resulted in a service that gained little traction.

But the electric tuk wasn't Sasiranga's endgame. Using the experience and skill he gained through the project, as well as the grant money, his research continued. He established Thermalit Industries and set his eyes on Sri Lanka's other most common mode of transport.

The Falcons Take Flight



The Peregrine

Inspired by the peregrine falcon: the world's fastest animal, Thermalit Industries created the Peregrine. A fully electric performance motorcycle made in Sri Lanka.

"We made it a mandate to source as many of the components and parts we could, locally. But of course if they weren't up to the standard we wanted, we had no choice but to import."

Sasiranga explained. We learned that the Peregrine is made with the utmost dedication to quality and craftsmanship, with no

compromise to the quality in components. "We wanted to make a bike that was on par with what's available in the international market in Europe for example, but was competitively priced in Sri Lanka," he said as we walked towards the Peregrine.

Seeing her for the first time, we could immediately notice a difference in craftsmanship. The long wheelbase and sleek design gave it a streamlined look and an aggressive stance. It truly looks unlike any other motorcycle found in the streets of Sri Lanka.

Turned out I wasn't alone in my view. Passersby never failed to stop and steal a glance or two at the Peregrine as we were out, taking pictures. It seems Thermalit hasn't failed in the design department.

A little art, a lot of engineering

Of course, the Peregrine isn't only



a good-looking motorcycle. It's also

well engineered. "Ride dynamics is something that we paid a lot of attention to when designing it," Sasiranga recounted. "We wanted the rider to have the confidence of a stable, balanced and comfortable ride."

Using complex simulations and other sorts of automotive and engineering whizbang, I could safely say that the team at Thermalit have been able to achieve just that. Although a die-hard petrolhead, I'm not much of a motorcycle guy. But I did get to ride in tandem.

At the rider's hand, the electric motor was smooth and responsive, with no noise or vibrations thanks to the lack of an engine. All we could hear was the futuristic hum of the electric motor powering up, a noise that was satisfying and exciting as the instant torque of the motor had us reaching the speed limit before I could blink twice.

Although I don't ride motorcycles, Kelum, who's usually behind the camera, does.

It only took him a moment before he was completely comfortable with being confident on how much he freed the throttle, a testament to how balanced the bike actually is. A couple of rounds in the motorcycle and he was grinning ear to ear.

But is it practical?

As good as the Peregrine might be to ride, it has to be functional and practical in order to combine

beyers to cash in. We turned to Sasiranga for answers.

"We had the bike undergo a rigorous testing period after building this one here," he revealed, resting his hand on the motorcycle in front of us. "We rode it in different road conditions and scenarios, and can confidently say that it can do everything you need in a motorcycle, with all the benefits of electric power."

What might these benefits be?

Well, the biggest one would be the benefit of not having to rely on fuel. Not only are electric motorcycles more efficient, relying on fossil fuels may not be the best course of action given the present day circumstances. Thermalit promises that all riders will have all the excitement of a performance bike, while having to only pay a cost of 80 cents per kilometre in Sri Lanka's Rupes.

Then there's all the savings that come from not having to do all the regular maintenance that you would normally have to do with a combustion engine. Saving both your time and money.

In fact, the Peregrine required very little maintenance. When you do, Thermalit is ready to provide all after-sales care necessary.

"That most of the parts are similar to a regular motorcycle anyway, so aside from the electric components, you could easily go to your nearest mechanic or mechanic to get any general repairs done."

Savings?

Of course, an electric bike doesn't come without its caveats. You will have to replace the battery once it starts to degrade. But Sasiranga assures you'll only need to worry about that once in every five to eight years or so. A service that Thermalit will provide for.

"But even then, the actual expenses you would have to bear with a traditional motorcycle over those years would still be far greater than you would have to bear," Sasiranga revealed.

Road ready

The Peregrine is a work of Sri Lankan engineering and design that left us wholly impressed. Enough to even sway a car guy like me to consider investing in one. When comparing the numbers, it simply makes sense. The Peregrine is fast, is great value for money, is a product of Sri Lankan ingenuity and most of all, is fun to ride.

Of course, not everyone needs a performance motorcycle. Some just need a practical commuter that is great value for money, but still fun and dynamic. Sasiranga has taken that into account as well, creating the Barbary: the Peregrine's less aggressive younger brother. So far, public interest on Thermalit's motorcycles are high and buyers have already reached out to Sasiranga, interested in purchasing a Peregrine of their own; a positive sign for the future of electric transportation in Sri Lanka.

"I want to change the mistaken opinion on EVs that Sri Lankans often have," Sasiranga commented. "Sri Lanka needs more production if we want to rise up from the current economic crisis that we have. There is no alternative. We are a local start-up who wants to make that happen."

Both the Peregrine and the Barbary are open for pre-orders through their official website thermalit.com.lk. Not only that, Sasiranga also wanted to extend an open invitation for anyone to come and see the result of his decade's worth of research for themselves.

"Come, see it for yourself and give it a test ride. If you decide for yourself that there is value in what we are offering, we assure you the best service possible with your purchase."

(Pix by Kelum Chamara)

The Barbary

Thermalit Barbary	1 Rs 595,000
Motor	1 4kW max
Range	1 120km
Top speed	1 80kmph
Type	1 3.5 x 17 inch
Advance payment	1 Rs 70,000
Delivery time	1 6 months after advance payment
Warranty	1 1 year, 3 years for battery

The Peregrine

Thermalit Peregrine	1 Rs 740,000
Motor	1 8kW max
Range	1 120km
Top speed	1 110kmph
Advance payment	1 Rs 70,000
Delivery time	1 6 months after advance payment
Type	1 6 x 17 inch
Warranty	1 1 year, 3 years for battery



DESIGN, PROTOTYPING AND TESTING OF A CUSTOM SHUTTLECOCK DISPENSER SYSTEM

LANKATRONICS (PVT) LTD.

[CLICK HERE TO VISIT WEBSITE.](#)

Overview:

This project involved the design, prototyping, and testing of a custom shuttlecock dispenser system utilizing two iris shutter mechanisms. The system was designed to efficiently manage shuttlecock dispensing with three operational states: Feed, Hold, and Release.

Objectives:

- Incorporate two iris shutters to control the dispensing mechanism effectively.
- Ensure the system is compact and portable.

Testing

- Prototyped a breadboard control circuit using two A4988 Stepper Motor Drivers to control the NEMA17 Stepper Motors.
- Programmed the system using Arduino IDE and tested for functionality,

Design Process:

- Solid Modelling and Motion Study conducted using SolidWorks .

Prototyping:

- Additive Manufacturing using PLA+ as the material

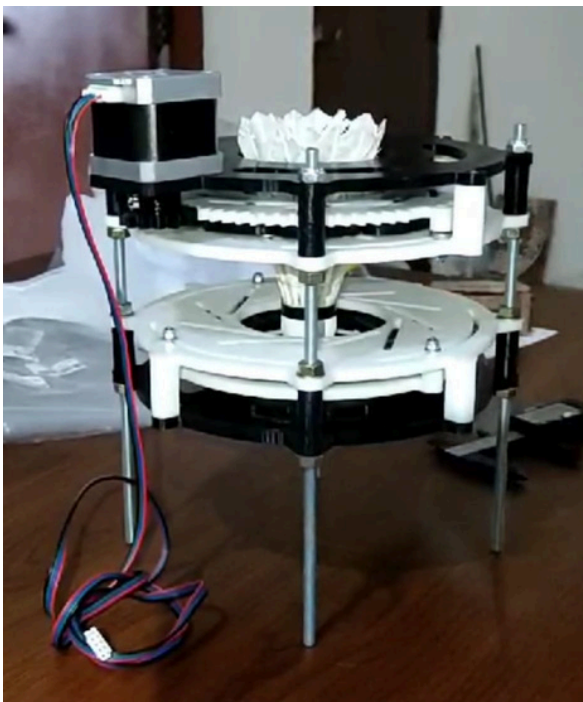


Figure 8: Final Assembled Prototype

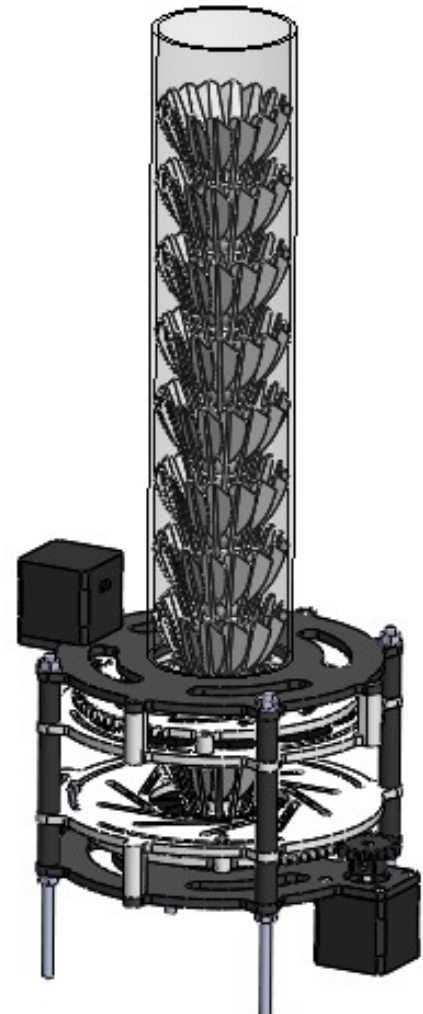


Figure 9: Final Design

DIY ROBOTIC PRODUCTS DEVELOPMENT FOR OPTIMUS ROBOTICS

LANKATRONICS (PVT) LTD.

[CLICK HERE TO VISIT WEBSITE.](#)

Overview:

This involved with the design and manufacturing of DIY robotic products under the brand Optimus Robotics, a subsidiary of Lankatronics (Pvt) Ltd. This main aim of this brand is to create accessible, user-friendly robotics kits for hobbyists and educators.

Objectives:

- Develop modular and customizable robotic kits that cater to various skill levels.
- Ensure an easy assembly with comprehensive instructions and support materials.
- Incorporate sustainable materials and components for eco-friendly production.

Design Process:

- Conducted market research and latest trends to identify user needs and preferences.
- Created CAD models for components, focusing on modularity and compatibility.
- Developed engaging instructional materials, including user manuals.

Manufacturing:

- Utilized Additive Manufacturing for prototype development and component fabrication.



Figure 10: Optimus Spiderbot

THE END